

Sri Lanka Institute of Information Technology



Requirement Analysis Activity
Information Technology Project (IT2080)
2025
Activity 01_ITP25_B4_76

Submitted by:

	Registration Number	Name with Initials (Surname first)	Email	Contact Phone Number
1.	IT23860032	Ambagahawaththa N.S.	it23860032@my.sliit.lk	0761224358
2.	IT23858398	Dilranga M.M.P.M	it23858398@my.sliit.lk	0756966802
3.	IT23857230	Damsara T.M.D.	it23857230@my.sliit.lk	0713534939
4.	IT23858848	Janeesha K.K.A.H.	it23858848@my.sliit.lk	0740754727
5.	IT23617100	Pathiranage S.S.	it23617100@my.sliit.lk	0717892110

Content

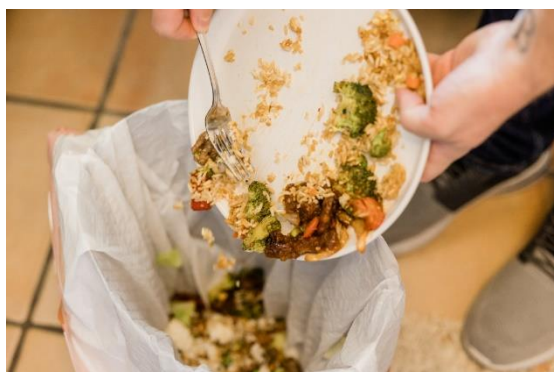
1. Brainstorm and identify a real-world problem that needs an IT solution. This can be a business requirement, an organization's requirement to improve the current process, or an innovative idea that solves a current issue.
2. Identify the users of the product and the benefits that they will get from the software system.
3. Present your ideas using a mind map
4. Prepare a list of user requirements for the features you have identified.
5. Suggest a suitable name for the project.

1.) Problem identification

In Sri Lanka and the entire world, plenty of food is wasted daily due to poor meal planning, improper portioning, and failure to utilize leftovers. Most homes, hostels, and restaurants cook meals without actually having an idea of how much food each individual requires. This leads to food wastage, unnecessary spending, and poor-quality nutritional balance.

Our web-based application The **PurePortion** addresses this problem by helping users plan meals accurately depending on the number of people, tracking nutritional values (proteins, vitamins, calories) and recycling leftovers. It gives wholesome, healthy meals and saves food wastage while saving time and money.

Sri Lanka generates
7,000 tonnes
 of solid waste per day – of
 which 56% is food waste.



Solution –

The system calculates the exact amount of ingredients needed for each dish based on the number of people, monitors nutritional content and suggests recipes using leftovers. It also provides grocery lists, health reports, and analytics to encourage efficient cooking and healthy eating habits.



100% Cost Effective Healthy meal

2.) Users and benefits

- **Households & Families**

- Plan daily meals with accurate ingredient measurements.
- Reduce food waste and save money.
- Access healthy and balanced meal plans.

- **Restaurants & Catering Services**

- Calculate ingredients for large groups without overcooking.
- Optimize ingredient usage to reduce cost.

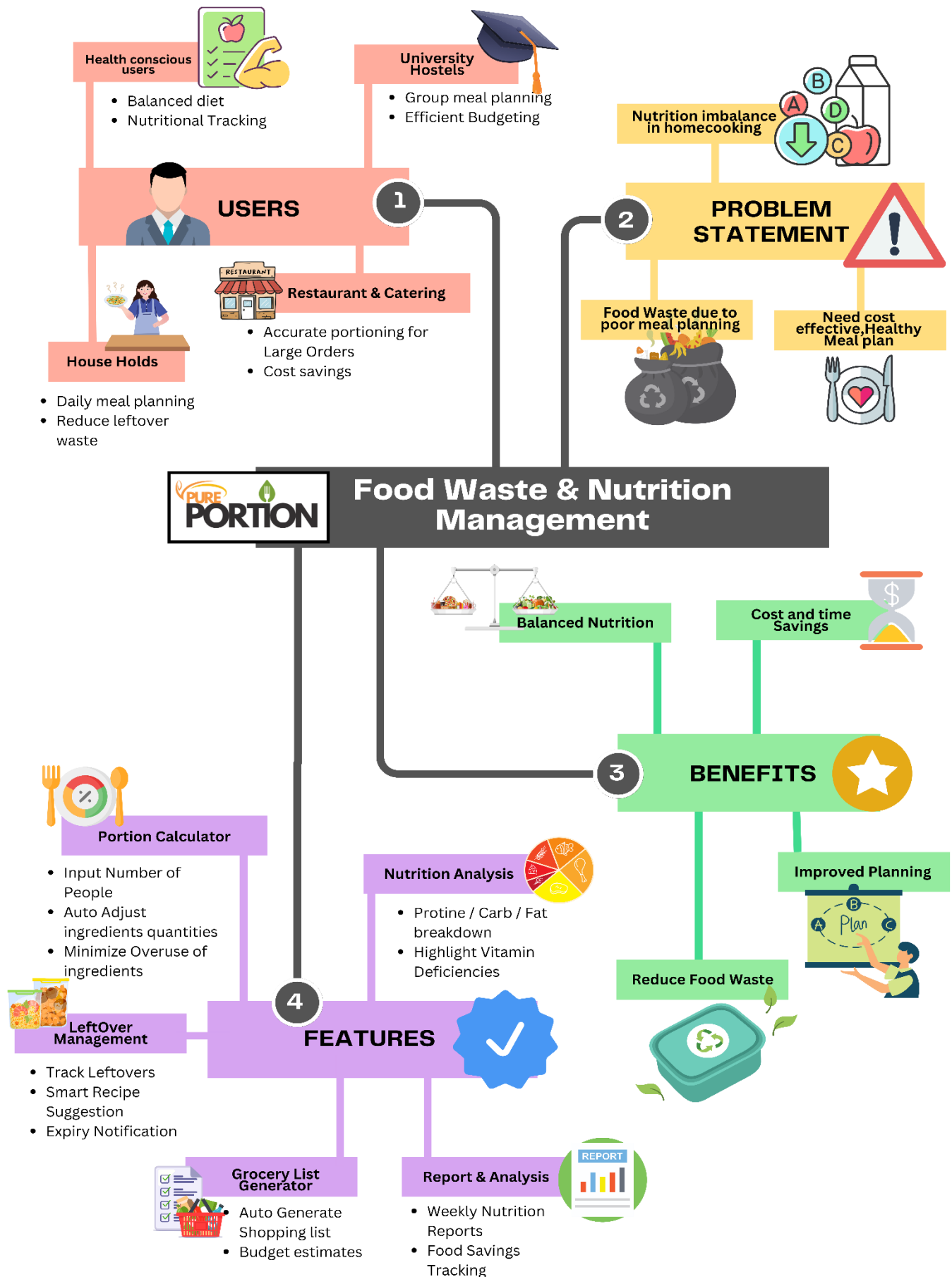
- **University Hostels & Canteens**

- Plan meals for multiple students at once.
- Generate bulk grocery lists automatically.

- **Health-Conscious Individuals**

- Track proteins, vitamins, and calories.
- Receive healthier meal recommendations.

3.) Present ideas using a mind map



4.) List of user main features

1. Portion Calculator

- Users must input the number of people, and the system will calculate ingredient amounts.
- Adjustments for special dietary needs (e.g., high protein).

2. Nutrition Analysis

- Display nutrition breakdown for selected meals.
- Highlight imbalances (e.g., too many carbs).

3. Leftover Management

- Add leftover ingredients with quantity and expiry date.
- Suggest recipes using leftover items.
- Send expiry reminders.

4. Grocery List

- Generate shopping lists from meal plans.
- Show cost estimates and quantity per item.

5. Recipe Management

- CRUD (Create, Read, Update, Delete) recipes with nutritional info.

6. Reports & Analytics

- Graphs showing weekly/monthly food saved.
- Nutrition reports (downloadable in PDF).

5.) Suggest suitable name for project

PurePortion – Smart Meal Planner & Food Waste Reduction System

